

RAKSHIT PAVAGADHI

✉ rakshit080905@gmail.com 🌐 Portfolio 📄 Rakshit Pavagadhi 🐙 rakshit-pavagadhi

EDUCATION

Pandit Deendayal Energy University

2023 – 2027

Bachelor of Technology (B.Tech.) in Computer Engineering — CGPA: 9.34

SKILLS

Programming Languages: C, C++, Python, Java, HTML, CSS, JavaScript, PHP

Frameworks & Libraries: Flask, React.js, Node.js, Express.js, Django, JSP, NumPy, Pandas, Scikit-learn, Librosa

AI / LLM: Prompt Engineering (Instruction, System, Grounding), Agentic Workflows, LLM APIs, Tool Calling, Embeddings

Databases: SQL, MySQL, PostgreSQL, MongoDB

Tools & Technologies: REST APIs, OAuth, Postman, Git, GitHub, SQLAlchemy, Bcrypt Authentication, Cloudinary

Soft Skills: Teamwork, Leadership, Problem-Solving, Communication

WORK EXPERIENCE

Infosys Springboard

Sept 2025 – Oct 2025

Intern

🐙 *GitHub Repo*

- Developed an end-to-end AI music composition system integrating a Random Forest ML pipeline with MusicGen to generate high-quality WAV outputs from natural language prompts.
- Built a secure Flask backend with JWT authentication, protected routes, session handling, and CORS-enabled APIs, establishing a scalable and modular server architecture for ML model serving and audio generation workflows.
- Collaborated in a team to design a hybrid composition engine that combined predicted genre outputs with structured MusicGen prompts, gaining hands-on experience in ML model deployment, API development, end-to-end pipeline design.

Projects

VoiceAI – Local Agent Pro | *Python, Whisper, LLMs, Groq, Streamlit*

🐙 *GitHub Repo*

- Engineered an agentic AI workflow integrating STT (Whisper), LLM reasoning (Llama 3 via Groq), and multi-tool execution with structured intent parsing for reliable tool-calling.
- Designed multi-intent command processing with contextual memory and prompt-engineered pipelines.
- Built a human-in-the-loop system with confirmation queues and sandboxed operations for safe, user-controlled automation.

Expense Management Platform | *MongoDB, Express.js, React, Node.js*

🐙 *GitHub Repo*

- Built a full-stack MERN expense platform with JWT authentication, Bcrypt hashing, and a scalable MongoDB schema.
- Implemented a greedy Min-Cash-Flow algorithm with dynamic splitting (equal, exact, percentage) and backend validation.
- Designed a glassmorphism UI with real-time dashboards, balance tracking, data visualization, and optimized API handling.

Empower Code Engine | *Next.js, TypeScript, FastAPI, Python, Monaco Editor*

🐙 *GitHub Repo*

- Built a browser-based Python execution platform that runs code via subprocesses, returning stdout, stderr, and exit codes.
- Engineered a strict 2-second timeout using Python's subprocess.run to gracefully terminate infinite loops.
- Implemented auto temp file cleanup, configured CORS-enabled REST APIs (loading, success, error, and timeout UI states).

Chat App | *React, Node.js, Express.js, Socket.io, MongoDB, Tailwind CSS*

🐙 *GitHub Repo*

- Built a full-stack real-time chat application with Socket.io-powered instant messaging, JWT authentication, Bcrypt password hashing, and persistent message history via MongoDB.
- Implemented real-time online/offline user status broadcasting to all connected clients and unseen message count tracking per conversation using Socket.io event management.
- Integrated Cloudinary, designed a responsive Tailwind CSS UI with React Context for global auth and chat state.

CERTIFICATIONS

Supervised Machine Learning: Regression and Classification [[Link](#)], Unsupervised Learning, Recommenders, Reinforcement Learning [[Link](#)], Advanced Learning Algorithms [[Link](#)]